Aarush Ram Anandh

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EXECUTIVE SUMMARY

Applied machine learning developer with real-time detection and interactive vision prototypes.
Doubled rare-class correct detection from 40% to 80%; built workflows for 20,000+ training inputs.
Python/C++ and sensor-data pipelines, including asthma detection at approximately 70% sensitivity.

EDUCATION

Ashoka University — B.Sc. (Hons.), Computer Science and Mathematics **Expected 2028**

RELEVANT EXPERIENCE

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Raised rare-class correct detection from **40% to 80%** (**+40 percentage points**) in an **RT-DETR** detection, classification and tracking pipeline for **18 waste-material classes**.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning for rare classes; evaluated detector failure modes through crop-based refinement.
- Helped set up a **near-infrared waste-segregation system** for visually similar plastics, including parts procurement and preliminary experiments.

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved latent-diffusion colour fidelity to $\Delta E_{00} < 1$, with approximately **4× better colour accuracy** than evaluated open-source baselines; results reported in the preprint below.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, an optimal-transport framework with control-conditioned probes, projections and slice aggregation; paper submitted to **JMLR**.

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Contributed to a wearable-data pipeline achieving approximately **70% sensitivity (recall)** for **asthma exacerbation detection** in study evaluation.
- Built sensor-data preprocessing, smoothing and change-point features for **191 patients**, with patient-wise train/test evaluation across five seeds.

SELECTED ENGINEERING PROJECTS

Shooting Stability Trainer **Sep 2025**
Computer Vision Prototype [GitHub]

- Prototyped webcam-based motion tracing with **OpenCV, Flask-SocketIO and JavaScript**; used HSV thresholding, contour filtering and centroid estimation to track a coloured marker.
- Built a browser interface with segmentation-mask feedback, adjustable HSV thresholds, two-point scale calibration, aim-centre selection and manual motion-trace recording.

CDR-Decomposition Pipeline **Jan 2026**
Vector Graphics / Training Data [GitHub]

- Built a **Python** pipeline to isolate structural die-line vectors from CorelDRAW packaging artwork, supporting generation of **20,000+ training inputs** for lab use.
- Automated headless **LibreOffice** CDR-to-SVG conversion, SVG/XML parsing, raster separation, luminance conversion and geometric filtering, retaining inspection outputs and metadata.

PUBLICATION

Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling (2026).
Public preprint: arXiv:2601.17259; colour-fidelity research described above.

TECHNICAL SKILLS

Programming: Python, C++, C, Bash. **ML / Vision:** PyTorch, TensorFlow, OpenCV, RT-DETR.
Prototyping: Flask-SocketIO, JavaScript, MATLAB/Simulink; detection, tracking, image processing.